

The following information applies to a four-lane urban freeway. $N=2$

Volume = 2,400 vph (in one direction)

Peak-hour factor = 0.90

Base free-flow speed = 60 mph

5% truck traffic in peak hour

12-ft lanes adjustment lane width, $f_{LW} = 0$

10-ft outside shoulders adjustment to total lateral clearance, $f_{TLC} = 0$

3 interchanges in 6 miles ramps/mile = 6 ramps/6 miles = 1

Level terrain; $E_T = 2$

No recreational vehicles

The level of service (LOS) for the freeway is most nearly:

- ☐ A. LOS A
- ☐ B. LOS B
- ☒ C. LOS C
- ☐ D. LOS D

③. ADJUST DEMAND VOLUME

$$V_P = \frac{V}{P_H F \times N \times f_{HV}}$$

$$f_{HV} = \frac{1}{1 + P_T (E_T - 1)}$$

$$= \frac{1}{1 + (0.05)(2 - 1)} = \frac{1}{1.05} = 0.95$$

$$V_P = \frac{2,400 \text{ vph}}{0.90 \times 2 \times 0.95}$$

$$V_P = 1403.50 \text{ pc/h/ln}$$

④. ESTIMATE SPEED DENSITY

$$D = \frac{V_P}{S}$$

$$D = \frac{1403.50 \text{ pc/h/ln}}{56.78 \text{ mph}}$$

$$D = 24.71 \text{ pc/mi/ln}$$

LOS C

②. ESTIMATE ADJUST CAPACITY

$$C = 2,000 + 10 \times (FFS - 50)$$

$$= 2,000 + 67.8$$

$$= 2067.8 \text{ pc/h/ln}$$

$$\begin{aligned} D \cdot FFS &= 2067.8 - f_{LW} - f_{TLC} - 3.22 \times TRD^{0.84} \\ &= 2067.8 - 0 - 0 - 3.22 \times 1^{0.84} \\ FFS &= 56.78 \text{ mph} \end{aligned}$$